Networked Constraint Expression: A Minimal Framework for Relational Discrepancy on Constrained Graphs

Abstract

We extend a scalar formulation of relational discrepancy to a networked setting and characterize the structural behaviors that arise under constrained interaction. Nodes are assigned a discrepancy variable defined as the difference between normalized observables, while edges encode interaction constraints. We introduce three operator classes—linear interaction, state-dependent interaction, and hard constraint—and show that they induce distinct regimes of system behavior. These include dispersal, throttling, retained accumulation, dynamic isolation, rerouting, and fragmentation. The results are structural: they follow from properties of constrained operators on graphs and do not depend on domain-specific interpretation or physical transport assumptions. The framework provides a minimal basis for analyzing how relational discrepancy is expressed, retained, and redistributed in constrained systems, as developed in prior scalar formulations of relational constraint expression (Castle, 2026a; Castle, 2026b).

1. Introduction

Scalar representations of system state often rely on observable signals that only partially reflect underlying conditions. Let a discrepancy variable capture the difference between local burden and expressed signal. While such a scalar can be informative, it does not specify how discrepancy is expressed across interacting units.

This paper provides a minimal extension: we embed scalar discrepancy in a graph and study how constrained interaction shapes its evolution. The goal is not to model any particular domain, but to identify structural behaviors that arise from operator form and topology.

2. Scalar Foundation

Let each node i carry two normalized observables:

- S_i : local pressure or burden
- R_i: expressed signal

Define relational discrepancy:

$$\Phi_i = R_i - S_i$$

This formulation follows prior work in which discrepancy variables were introduced as reduced-form representations of system constraint expression (Castle, 2026a; Castle, 2026b).

Φ_i is not a physical quantity. It is a reduced-form representation of mismatch between local conditions and their expression.

$q_{i,t+1} = q_{i,t} + inflow_{i,t} - clearance_{i,t}$

This term captures persistence of unresolved local burden across time.

3. Graph Embedding

Let $G=(V, E)$ be a graph with:

- nodes $i \in V$
- edges $ij \in E$
- weights $W_{ij} \geq 0$, representing interaction strength

Define degree matrix D and Laplacian:

$$L = D - W$$

The graph represents a constraint on interaction. It is not a physical transport medium.

4. Operator Classes

We define three operator classes governing the evolution of Φ.

4.1 Linear Interaction Operator

$$\Phi_{t+1} = \Phi_t - \eta L \Phi_t - \gamma \Phi_t$$

Where:

- $\eta > 0$: interaction strength
- $\gamma > 0$: local damping

This operator defines local interaction and decay without explicit constraints. The form corresponds to standard graph Laplacian dynamics used in diffusion and consensus processes (Chung, 1997).

4.2 State-Dependent Interaction

$$W_{ij,t} = W_{ij} \cdot f(\mid \Phi i - \Phi j \mid)$$

where $f(\cdot)$ is a monotone function.

Interpretation:

- interaction strength varies with local discrepancy

4.3 Hard Constraint Operator

$$\widetilde{W}_{ij,t} = W_{ij} O_{ij,t} F_{ij,t}$$

$$F_{ij,t} = f(H_{ij,t}, A_{j,t})$$

where:

- $O_{ij,t} \in [0,1]$ = physical openness
- $F_{ij,t} \in [0,1]$ = functional transmissibility

This defines a state-dependent effective graph G_t induced by Φ_t.

Edges may be conditionally removed from the effective graph.

Under state-dependent and hard constraint operators, the effective interaction graph becomes a function of the system state.

4.4 Directed Interaction Operator

In prior formulations of scalar interaction dynamics (Castle, 2026a; Castle, 2026b), local interaction was expressed through gradient-based influence functions operating on a scalar state variable m_i, representing local system condition. In that setting, directional influence was modeled as:

$$P_{i,t} = \lambda \sum_j W_{ij} max(0; m_{j,t} - m_{i,t})$$

Where influence is asymmetric and arises only when neighboring states exceed the local state.

In the present framework, the discrepancy variable Φ replaces m, as defined in the scalar framework (Castle, 2026a; Castle, 2026b), yielding:

$$P_{i,t} = \lambda \sum_j W_{ij} max(0; \Phi_{j,t} - \Phi_{i,t})$$

The state update is then defined as:

$$\Phi_{t+1} = \Phi_t + P_t - \gamma \Phi_t$$

This operator introduces **directional interaction based on local discrepancy gradients**, in contrast to the symmetric interaction induced by the Laplacian operator.

Interpretation

- Influence flows from nodes with higher discrepancy to nodes with lower discrepancy
- Interaction is local and edge-mediated
- Directionality arises from the ordering of node states rather than from imposed structure

Relation to Prior Formulations

The directed operator generalizes earlier gradient-based interaction rules defined over scalar state variables by substituting the discrepancy variable Φ for m. This preserves the original directional influence structure while embedding it in a relational discrepancy framework.

Role in the Operator Set

The directed interaction operator complements the linear (Laplacian) operator:

- **Laplacian operator**: symmetric interaction, diffusion-like behavior
- **Directed operator**: asymmetric interaction, gradient-driven propagation

Both operate under the same graph structure and may be used independently or in combination, depending on the interaction regime.

5. Structural Behaviors

The following behaviors arise from the interaction between operator structure and graph topology.

5.1 Dispersal

Under linear interaction:

- discrepancy spreads across the graph
- decay rate is governed by the operator spectrum

5.2 Local Propagation Ordering

Under local interaction operators, changes in node state propagate across the graph through successive neighbor interactions.

Because interaction is edge-mediated, the influence of a localized perturbation spreads in order of graph distance, with adjacent nodes affected prior to more distant nodes.

Node states evolve through constrained interaction with neighbors.

This produces a form of propagation ordering in which the temporal response of nodes reflects their shortest-path distance from the source of perturbation.

Statement:
Under local interaction operators on a connected graph, the effect of a localized discrepancy change propagates outward in graph-distance order.

Propagation ordering arises under both symmetric (Laplacian) and directional (gradient-based) interaction operators.

5.2 Throttled Interaction

Under state-dependent weights:

- interaction weakens across high-gradient edges
- local persistence increases

5.3 Retained Accumulation

Under edge removal with persistent inflow:

- nodes upstream of constraints accumulate discrepancy
- local plateaus form

Statement:
If net inflow exceeds effective outflow due to constraint, accumulation occurs.

5.4 Dynamic Isolation

When dominant interaction paths are removed:

- node dynamics decouple from neighbors

Statement:
Removal of primary interaction paths induces conditional independence between node states.

5.5 Rerouting

Under partial constraint:

- discrepancy redistributes along alternative paths

5.6 Fragmentation

Under sufficient edge removal:

- the graph partitions into disconnected components consistent with standard results in graph theory (Chung, 1997).
- subgraphs evolve independently

Statement:
Constraint-induced edge removal can produce disconnected subgraphs with independent dynamics.

6. Regime Structure

System behavior can be classified by operator state.

Regime I: Linear Regime

- no active constraints
- smooth dispersal
- amplitude-invariant decay

Regime II: Constrained Regime

- partial reduction in interaction strength
- slower propagation
- increased persistence

Regime III: Binding Constraint Regime

- edge removal
- accumulation, isolation, and rerouting

Regime IV: Fragmented Regime

- multiple constraints active

- disconnected components
- independent evolution

Key observation:
Regimes are induced by changes in effective graph connectivity and interaction strength.

7. Local Interaction Structure

Node dynamics are governed by local interaction operators acting on neighboring discrepancies.

Influence between nodes is:

- **local**: restricted to edges in the graph
- **mediated**: determined by interaction weights
- **state-dependent**: modified by constraint activation

No global interaction is assumed. All system behavior arises from repeated local updates.

Under constraint, interaction may become:

- **asymmetric**, when transmissibility differs across edges
- **intermittent**, when edges are conditionally removed

Thus, cross-node influence is not an independent mechanism, but a consequence of local interaction structure and operator form.

8. General Properties

The following properties hold generically for local interaction operators on constrained graphs:

- propagation follows graph-local interaction
- accumulation arises from asymmetric flow
- rerouting occurs under partial constraint
- isolation follows removal of dominant paths
- fragmentation follows sufficient edge removal

These properties are mathematical and do not depend on domain-specific interpretation.

9. Interpretation Boundary

This framework describes how relational discrepancy may be expressed across constrained interactions.

It does not imply:

- physical transport
- conservation laws
- domain-specific flow mechanisms

All operators act on a reduced-form representation.

10. Conclusion

A scalar discrepancy variable can be embedded in a graph to study how constrained interaction shapes its evolution.

These results extend the scalar discrepancy formulation by specifying how such discrepancy is expressed under constrained interaction.

Three operator classes—linear, state-dependent, and hard constraint—produce distinct structural behaviors, including dispersal, accumulation, isolation, rerouting, and fragmentation.

These behaviors arise from general properties of constrained operators on graphs. The framework provides a minimal, domain-independent basis for analyzing how discrepancy is expressed, retained, and redistributed in interacting systems.

References

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